

plan

A Practical Evaluation of Oversmoothing and Oversquashing in Graph Neural Networks

Malaz Natouf

Learning on Graphs
RWTH Aachen University
Aachen, Germany

Abstract: Deep GNN depth degradation is often explained through oversmoothing or oversquashing, but the practical relevance of these mechanisms depends on the task, graph, optimization regime, and intervention. This thesis plan combines broad real-world screening with controlled experiments and uses predictive performance together with representation diagnostics and intervention-based checks. The oversmoothing block is supported by audited real-world, controlled-homophily, cSBM, learning-rate, and PairNorm-ablation experiments. Complementing the oversmoothing analysis, the oversquashing experiments investigate when long-range information loss can be attributed to structural bottlenecks and whether such bottlenecks also translate into predictive degradation on controlled and real-world graphs. The goal of the final thesis is a cautious, reproducible account of when oversmoothing and oversquashing meaningfully explain deep-GNN degradation and when optimization, topology, task structure, or dataset properties provide stronger alternative explanations.

1 Introduction

1.1 Motivation

Increasing GNN depth enlarges the receptive field, but deeper message passing can also degrade predictive performance. This thesis investigates when such degradation is associated with oversmoothing, when normalization can mitigate it, and how these observations relate to optimization and graph structure. The intended contribution is an empirical diagnosis rather than a universal claim: performance, representation geometry, optimization sensitivity, and controlled graph interventions are evaluated together.

1.2 Research Questions

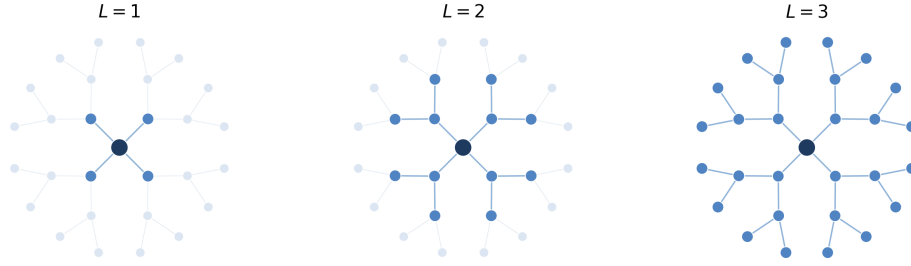
- RQ1.** How does increasing GNN depth affect predictive performance across real-world and controlled graph-learning tasks?
- RQ2.** When is depth-related performance degradation accompanied by oversmoothing-related representation collapse, and when does PairNorm mitigate both?
- RQ3.** How do source-target distance and structural bottlenecks affect long-range information propagation, and how do normalization and structural interventions alter this behavior?
- RQ4.** Under which conditions are oversmoothing and oversquashing useful explanations for deep-GNN failure, and when are optimization, task, or dataset properties more plausible explanations?

2 Background

2.1 Message Passing and Receptive Fields

Message passing updates a node representation by aggregating information from its neighborhood [Gi17]. Standard realizations include GCN [KW17] and GraphSAGE [HYL17]. Each additional layer expands the graph-distance range over which information can influence a target representation.

Message-Passing Depth and Receptive Field



Increasing depth expands the set of nodes that can theoretically influence the target node.

Figure 1: Receptive-field growth under repeated message passing. Increasing depth expands the set of nodes whose information can influence a target representation.

2.2 Depth versus Hidden Dimension

Depth and hidden dimension play different roles. Depth controls how many message-passing steps are available, while hidden width controls the capacity of the intermediate representation. This distinction is important for both oversmoothing and oversquashing analyses.

Depth versus Hidden Width

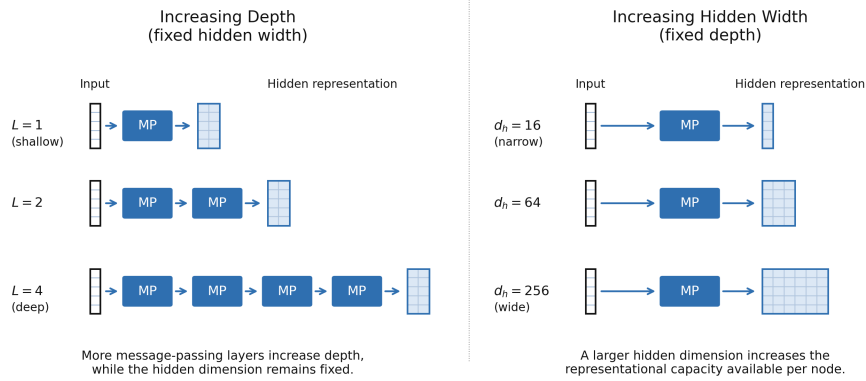


Figure 2: Conceptual distinction between depth and hidden width in message-passing GNNs. Depth changes propagation distance, whereas width changes representation capacity.

2.3 Homophily and Heterophily

Graph-learning behavior depends on the relation between topology and labels. Homophily is useful for characterizing this relation, but it does not by itself determine task difficulty or model behavior [Pl23; Zh20].

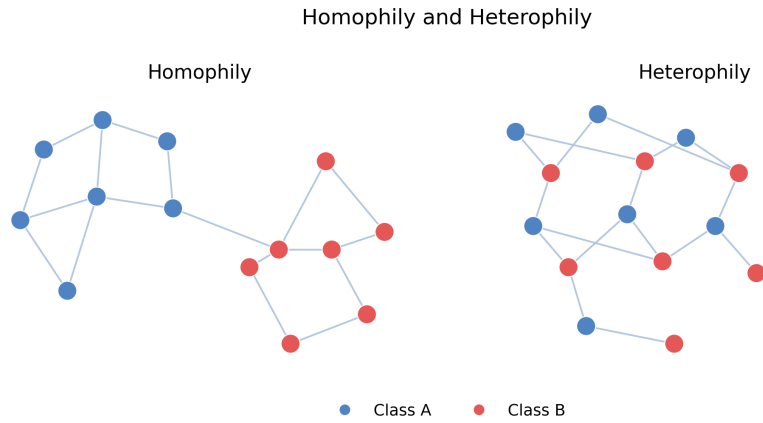


Figure 3: Schematic contrast between homophilous and heterophilous graph structure. The controlled experiments later vary label-edge alignment directly rather than relying only on unrelated real-world datasets.

2.4 Oversmoothing

Repeated graph convolution has been linked to Laplacian smoothing and progressive loss of feature variation [LHW18]. The broader literature emphasizes that multiple definitions and diagnostics coexist and that representation collapse should not automatically be equated with predictive failure [RBM23]. In this thesis, normalized Dirichlet energy (NDE) and effective rank are primary diagnostics, with cosine-based measures used as supporting evidence.

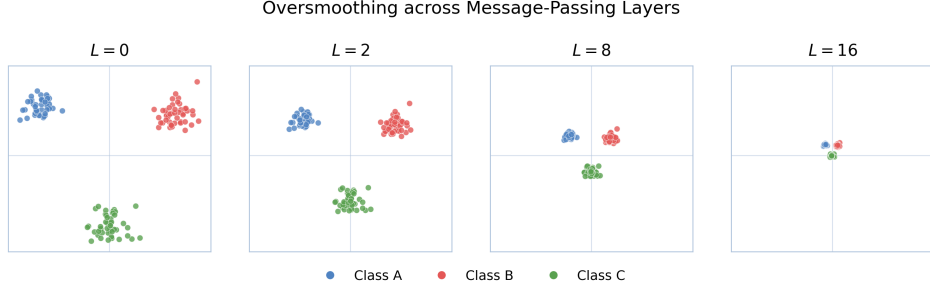


Figure 4: Conceptual illustration of oversmoothing: repeated aggregation can reduce discriminative variation between node representations.

2.5 PairNorm

PairNorm was proposed as a normalization method for counteracting oversmoothing by controlling representation geometry [ZA20]. The later center-only and scale-only experiments are thesis-specific diagnostic ablations and are not presented as variants from the original PairNorm paper.

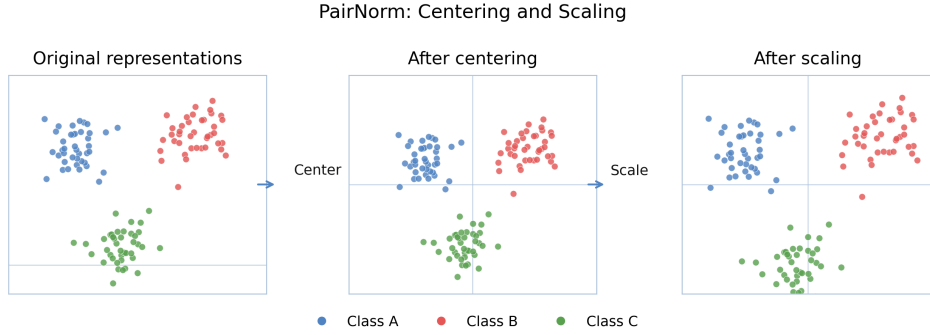


Figure 5: Conceptual effect of PairNorm on node-representation geometry.

3 Experimental Methodology

3.1 Real-World Datasets

The broad screen covers 15 real-world node-classification datasets. Edge counts are unique undirected pairs. The final column of Table 1 records the benchmark/source corresponding to the employed dataset version.

Table 1: Dataset overview. Edge counts denote unique undirected pairs; the final column identifies the benchmark/source used for the employed dataset version.

Dataset	Nodes	Edges	Features	Classes	Edge homophily h	Loader	Source / reference
Cora	2,708	5,278	1,433	7	0.810	Planetoid	[YCS16]
CiteSeer	3,327	4,552	3,703	6	0.736	Planetoid	[YCS16]
PubMed	19,717	44,324	500	3	0.802	Planetoid	[YCS16]
Roman-Empire	22,662	32,927	300	18	0.047	HeterophilousGraphDataset	[PI23]
Amazon-Photo	7,650	119,081	745	8	0.827	Amazon	[Sh18]
Amazon-Computers	13,752	245,861	767	10	0.777	Amazon	[Sh18]
Coauthor-CS	18,333	81,894	6,805	15	0.808	Coauthor	[Sh18]
Coauthor-Physics	34,493	247,962	8,415	5	0.931	Coauthor	[Sh18]
Actor	7,600	26,752	932	5	0.219	Actor	[Pe20]
Chameleon	2,277	31,421	2,325	5	0.235	WikipediaNetwork	[Pe20; RAS21]
Squirrel	5,201	198,493	2,089	5	0.224	WikipediaNetwork	[Pe20; RAS21]
Cornell	183	280	1,703	5	0.131	WebKB	[Pe20]
Texas	183	295	1,703	5	0.108	WebKB	[Pe20]
Wisconsin	251	466	1,703	5	0.196	WebKB	[Pe20]
Amazon-Ratings	24,492	93,050	300	5	0.380	HeterophilousGraphDataset	[PI23]

Note. Features denotes the node-feature dimensionality; Classes is the number of target classes; edge homophily h is the raw fraction of edges whose endpoints share the same class label. Loader denotes the PyTorch Geometric dataset loader/class used for the employed dataset version.

3.2 Evaluation Protocol

The evaluation follows three rules throughout the empirical plan:

- checkpoints are selected using validation accuracy;
- hyperparameter/model selection uses validation performance, with test accuracy reported only after selection;
- matched hyperparameters are used where the purpose is a mechanism comparison rather than a best-achievable configuration comparison.

3.3 Oversmoothing Diagnostics

The primary representation diagnostics are normalized Dirichlet energy and effective rank/effective-rank ratio. Pairwise and edge-based cosine measures provide supporting geometric evidence. Representation diversity is treated as a diagnostic, not as a sufficient condition for predictive performance.

4 Broad Real-World Screening

4.1 Experimental Design

The broad screen first establishes whether depth degradation is a general or heterogeneous phenomenon across real-world tasks. Phase 1 uses depths $L \in \{2, 4\}$; Phase 2 uses $L \in \{8, 16, 32\}$. Candidate families include GCN, GAT, GraphSAGE, and corresponding PairNorm variants where applicable. Configurations are selected by mean validation accuracy within each comparison block.

Table 2: Broad real-world screening across shallow and deeper phases. Phase 1 uses $L \in \{2, 4\}$ and Phase 2 uses $L \in \{8, 16, 32\}$. Each cell reports the validation-selected model, depth, hidden dimension, and test accuracy.

Dataset	Phase 1 baseline	Phase 1 full	Phase 2 baseline	Phase 2 full
Cora	GAT	GAT	GCN	GCN
	L4, H128	L4, H128	L8, H64	L8, H64
	79.50%	79.50%	$76.70 \pm 0.85\%$	$76.70 \pm 0.85\%$
CiteSeer	GraphSAGE	GraphSAGE	GCN	GCN
	L2, H128	L2, H128	L8, H128	L8, H128
	70.60%	70.60%	$63.27 \pm 0.31\%$	$63.27 \pm 0.31\%$
PubMed	GraphSAGE	GAT+PN	GAT	GAT+PN
	L2, H64	L4, H128	L8, H128	L8, H128
	77.20%	77.30%	$77.57 \pm 1.38\%$	$76.47 \pm 0.95\%$
Roman-Empire	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE+PN
	L4, H128	L4, H128	L8, H128	L8, H128
	$79.37 \pm 0.58\%$	$79.42 \pm 0.51\%$	$68.06 \pm 0.67\%$	$71.62 \pm 0.78\%$
Amazon-Photo	GAT	GAT	GCN	GraphSAGE+PN
	L2, H128	L2, H128	L8, H64	L8, H128
	$91.51 \pm 0.57\%$	$91.63 \pm 0.60\%$	$82.58 \pm 2.57\%$	$87.07 \pm 1.30\%$
Amazon-Computers	GCN	GCN	GCN	GraphSAGE+PN
	L2, H128	L2, H128	L8, H64	L8, H128
	$83.71 \pm 0.79\%$	$83.74 \pm 0.58\%$	$56.63 \pm 9.74\%$	$82.37 \pm 1.47\%$
Coauthor-CS	GCN	GCN	GCN	GCN
	L2, H128	L2, H128	L8, H64	L8, H64
	$91.22 \pm 0.69\%$	$91.22 \pm 0.69\%$	$86.17 \pm 0.92\%$	$86.09 \pm 0.90\%$
Coauthor-Physics	GCN	GCN	GCN	GCN
	L4, H64	L2, H64	L8, H64	L8, H64
	$92.83 \pm 0.29\%$	$93.20 \pm 0.54\%$	$92.30 \pm 0.56\%$	$92.21 \pm 0.49\%$
Actor	GraphSAGE	GraphSAGE	GCN	GraphSAGE+PN
	L4, H64	L4, H64	L8, H64	L8, H128
	$34.49 \pm 1.11\%$	$34.39 \pm 1.15\%$	$26.40 \pm 1.25\%$	$32.87 \pm 0.60\%$
Chameleon	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE
	L2, H64	L2, H64	L8, H64	L8, H64
	$49.82 \pm 2.32\%$	$49.82 \pm 2.32\%$	$42.83 \pm 2.80\%$	$42.57 \pm 2.74\%$
Squirrel	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE+PN
	L2, H64	L2, H64	L8, H128	L16, H64
	$36.95 \pm 2.27\%$	$36.95 \pm 2.27\%$	$31.99 \pm 1.65\%$	$31.76 \pm 0.85\%$
Cornell	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE+PN
	L2, H128	L2, H128	L8, H64	L8, H64
	$71.08 \pm 5.56\%$	$71.08 \pm 5.56\%$	$56.22 \pm 6.96\%$	$52.43 \pm 12.30\%$
Texas	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE
	L2, H128	L2, H128	L8, H128	L8, H128
	$78.65 \pm 5.76\%$	$78.65 \pm 5.76\%$	$65.68 \pm 7.21\%$	$64.86 \pm 7.32\%$
Wisconsin	GraphSAGE	GraphSAGE	GraphSAGE	GraphSAGE
	L2, H128	L2, H128	L8, H128	L8, H128
	$80.20 \pm 4.66\%$	$80.78 \pm 4.51\%$	$66.67 \pm 7.22\%$	$64.12 \pm 7.28\%$
Amazon-Ratings	GraphSAGE	GraphSAGE+PN	GAT	GraphSAGE+PN
	L4, H128	L4, H128	L8, H128	L8, H128
	$45.76 \pm 0.41\%$	$50.10 \pm 0.41\%$	$42.96 \pm 0.22\%$	$48.00 \pm 0.61\%$

PN = PairNorm. Selection is validation based; test accuracy is reported after selection.

4.2 Results

The broad screen is used to identify representative cases and to avoid building the thesis around a single dataset. The current result is heterogeneous: depth hurts strongly on some datasets but much less on others, and PairNorm is not uniformly beneficial.

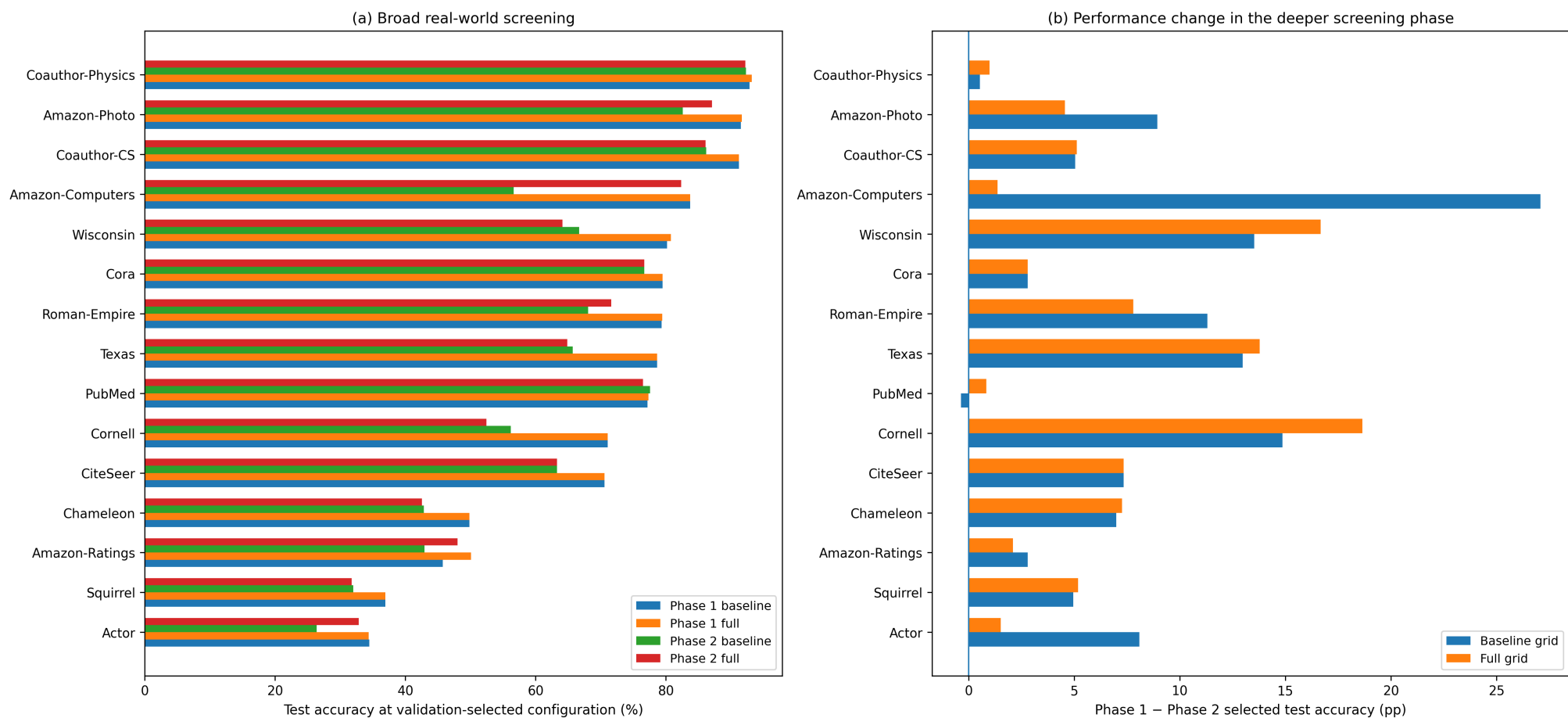


Figure 7a: Broad real-world screening: validation-selected predictive performance across depth. Configuration selection is based on validation performance; test performance is reported afterward.

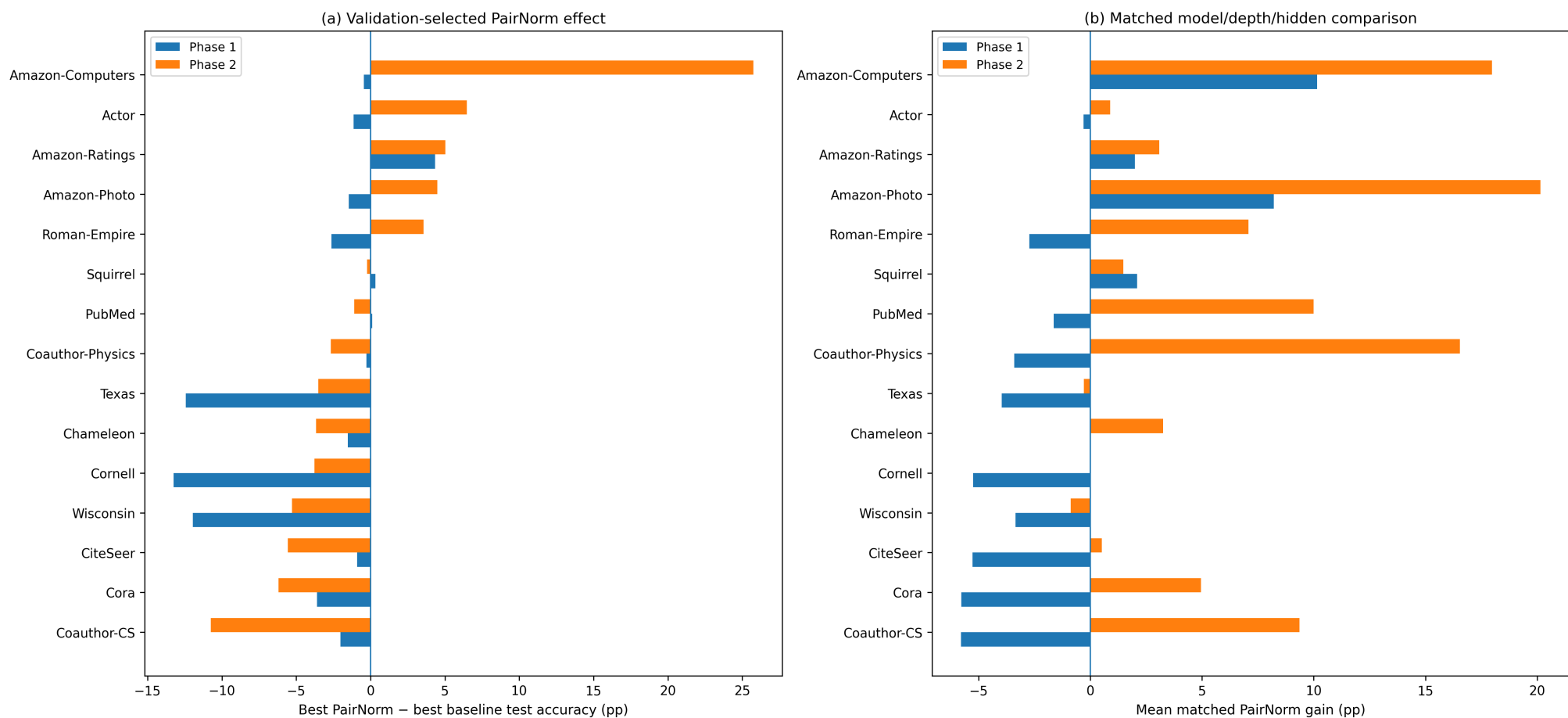


Figure 7b: Broad real-world screening: PairNorm effects under validation-selected and matched comparisons. The current effects are heterogeneous rather than uniformly positive.

5 Selected Real-World Cases

Four datasets are selected for more controlled comparison: Amazon-Ratings, Squirrel, Roman-Empire, and Actor. The design compares GraphSAGE with GraphSAGE+PairNorm at $L = 4$ and $L = 8$, hidden dimension 128, 800 training epochs, and 10 identical splits. The learning-rate grid is $\{0.0003, 0.001, 0.003, 0.005, 0.01, 0.03\}$.

Table 3: Selected-four real-world design. The cases deliberately provide contrasting PairNorm outcomes under a common depth pair and training protocol.

Dataset	Selection role	Models	Depths	Hidden	Splits	Epochs
Amazon-Ratings	Contrasting case with reduced depth degradation under PairNorm	GraphSAGE / GraphSAGE+PN	L = 4, 8	128	10 identical splits	800
Squirrel	Contrasting case with reduced depth degradation under PairNorm	GraphSAGE / GraphSAGE+PN	L = 4, 8	128	10 identical splits	800
Roman-Empire	Counterexample with increased depth degradation under PairNorm	GraphSAGE / GraphSAGE+PN	L = 4, 8	128	10 identical splits	800
Actor	Counterexample despite stronger representation-diversity diagnostics	GraphSAGE / GraphSAGE+PN	L = 4, 8	128	10 identical splits	800

5.1 Depth Interaction

Here, Acc_{L4} and Acc_{L8} denote the test accuracy at the validation-selected checkpoint for the same model trained with four and eight message-passing layers, respectively. The depth loss is defined as

$$\text{DepthLoss} = \text{Acc}_{L4} - \text{Acc}_{L8}.$$

Thus, a positive depth loss means that the eight-layer model performs worse than its four-layer counterpart.

For the matched PairNorm comparison, GS_{L4} and GS_{L8} denote the GraphSAGE test accuracies at depths 4 and 8, while PN_{L4} and PN_{L8} denote the corresponding GraphSAGE+PairNorm accuracies. All four quantities are compared at the same matched learning rate. The difference-in-differences (DiD) is

$$\text{DiD} = (\text{PN}_{L8} - \text{PN}_{L4}) - (\text{GS}_{L8} - \text{GS}_{L4}).$$

Equivalently, $\text{DiD} = \text{DepthLoss}_{\text{GS}} - \text{DepthLoss}_{\text{PN}}$. A positive DiD therefore means that PairNorm reduces depth-related degradation relative to GraphSAGE; a negative DiD means that it increases the degradation under the matched learning rate.

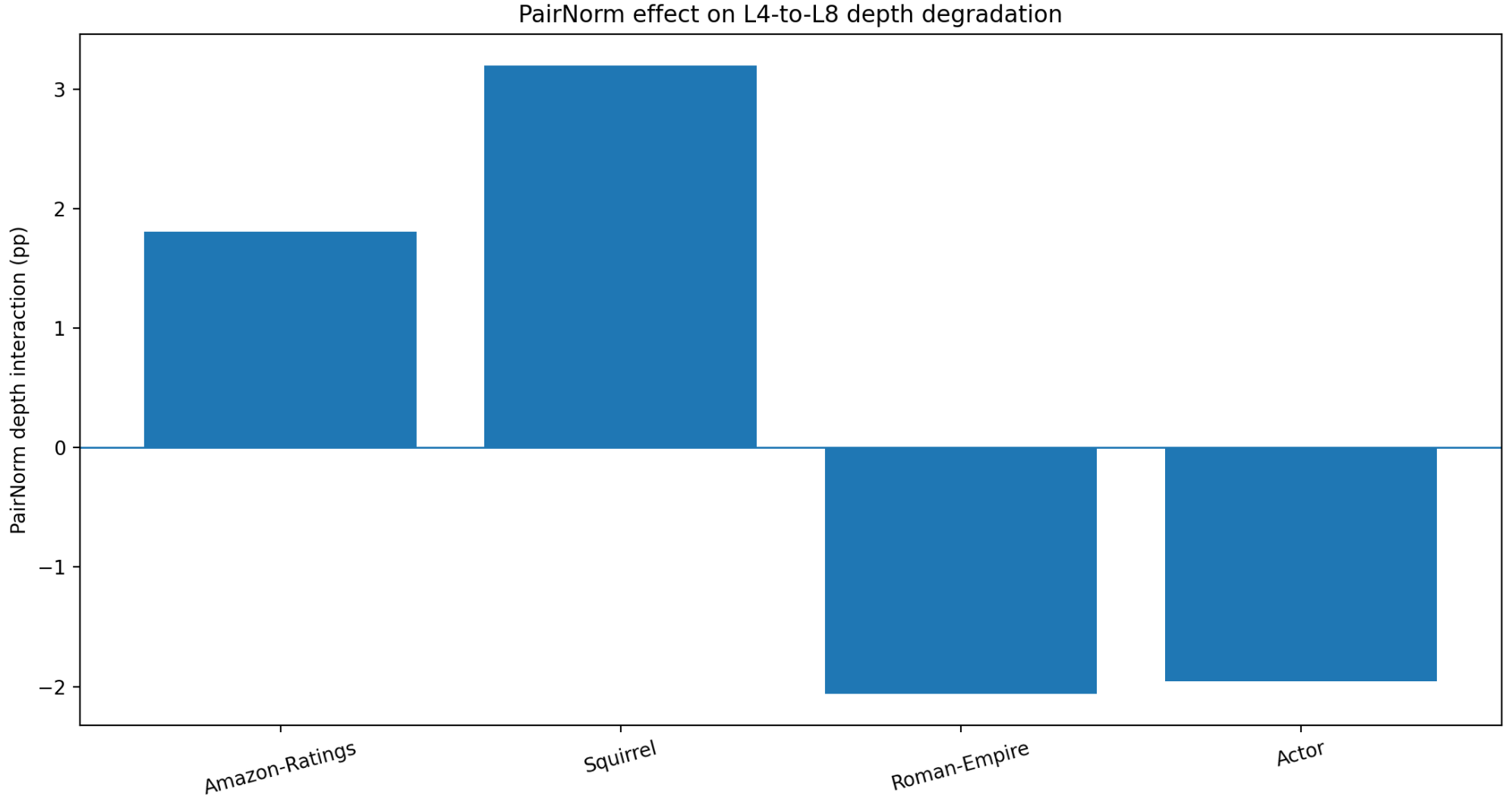


Figure 8: Selected real-world cases: matched depth interaction for GraphSAGE versus GraphSAGE+PairNorm. Positive values indicate reduced depth degradation under PairNorm.

Table 4: Matched selected-four results.

Dataset	LR	GS depth loss	PN depth loss	DiD	Holm p	Cohen d_z	NDE ratio	Rank ratio
Amazon-Ratings	0.003	2.734	0.928	+1.806	0.005859	+2.316	2.31×	3.63×
Squirrel	0.03	4.092	0.893	+3.199	0.005859	+2.977	3.14×	2.30×
Roman-Empire	0.005	0.184	2.243	-2.060	0.005859	-2.001	1.61×	0.99×
Actor	0.0003	1.178	3.132	-1.954	0.011719	-1.291	1.47×	2.30×

Note. LR = learning rate; GS = GraphSAGE; PN = GraphSAGE+PairNorm. Depth loss is $\text{Acc}_{L4} - \text{Acc}_{L8}$ in percentage points. DiD = matched difference-in-differences depth interaction; Holm p = Holm-adjusted p -value for the paired interaction test; Cohen d_z = paired standardized effect size. NDE and Rank ratio are PairNorm-to-GraphSAGE ratios at $L = 8$.

The selected cases deliberately include both directions: Amazon-Ratings and Squirrel show reduced depth degradation under PairNorm, while Roman-Empire and Actor show increased depth degradation. The effect is therefore conditional rather than universal.

5.2 Representation Diagnostics

Figure 9 tests whether improved representation diversity is sufficient to explain the performance interaction. It is not: Actor is an important counterexample, because PairNorm raises both NDE and the effective-rank ratio while worsening the depth interaction.

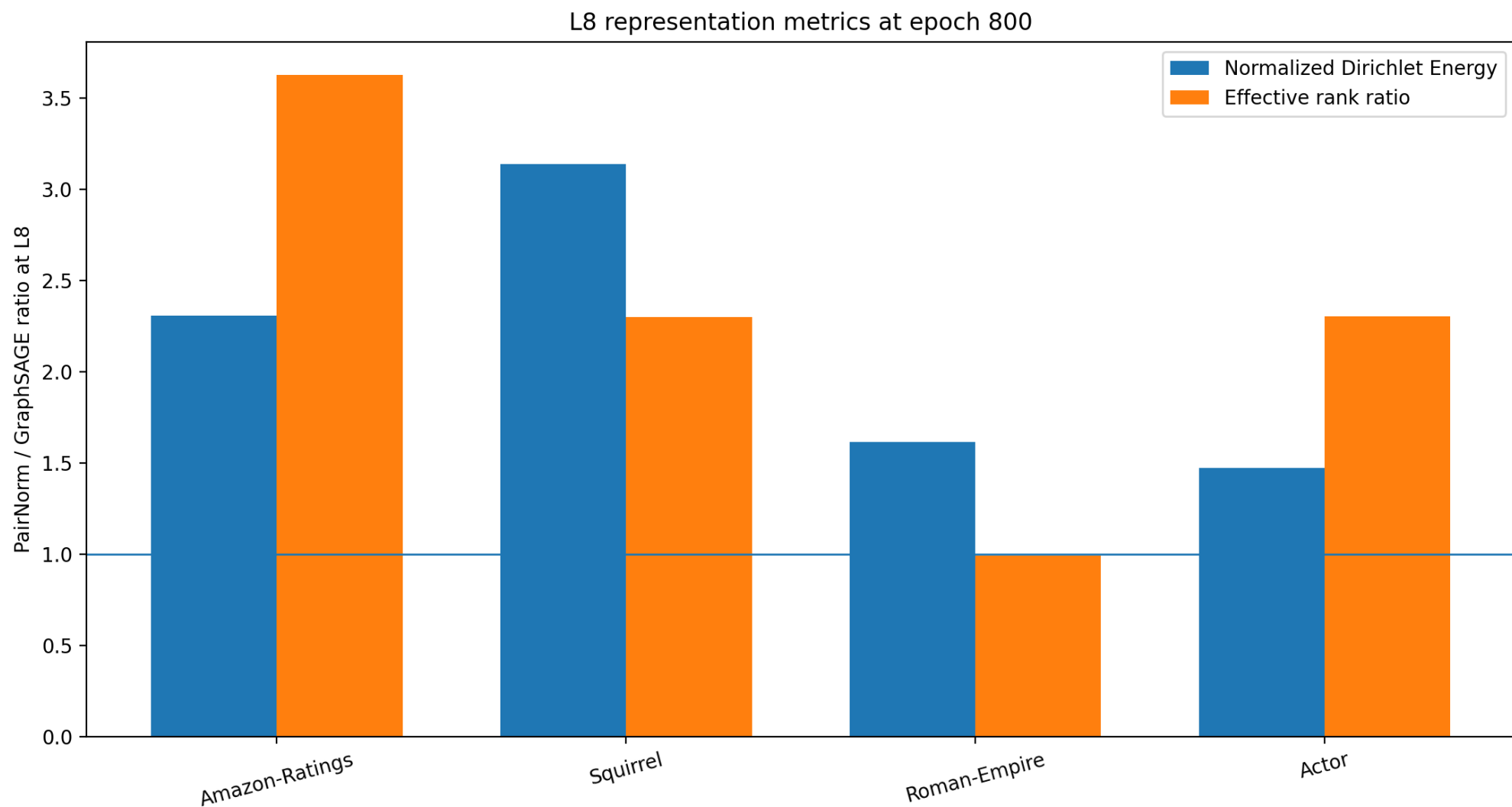


Figure 9: Relative representation-diversity changes at $L = 8$ under matched learning rates. Bars show the PairNorm-to-GraphSAGE ratio for normalized Dirichlet energy and for the effective-rank ratio. Values above one indicate larger representation diversity under PairNorm.

6 Controlled Homophily

6.1 Controlled Graph Construction

To reduce dataset-level confounding, controlled graphs are derived from PubMed and Roman-Empire. Node features and labels are retained while topology is modified by degree-preserving double-edge swaps on a simple undirected graph. The edge count and complete degree sequence are preserved. Raw edge-homophily targets are $h \in \{0.1, 0.5, 0.9\}$; adjusted homophily is recorded only as a diagnostic. The final graph is required to remain connected. The low-label split uses 5% train, 10% validation, and 85% test.

Table 5: Controlled-homophily factorial design.

Factor	Setting
Dataset families	PubMed-derived; Roman-Empire-derived
Target homophily	0.1, 0.5, 0.9
Models	GraphSAGE, GraphSAGE + PairNorm
Depths	7, 8
Hidden dimension	128
Graph seeds	1, 2, 3, 4, 5
Initialization seeds	11, 29, 47, 65, 83
Split index	0
Training epochs	2400
Observed factorial cells	600
Expected factorial cells	600
Missing factorial cells	0

6.2 Endpoint Results

The controlled experiment asks whether the relationship between depth, representation collapse, and PairNorm changes as label-edge alignment is modified while node features and labels remain fixed.

Controlled Homophily: Depth and PairNorm

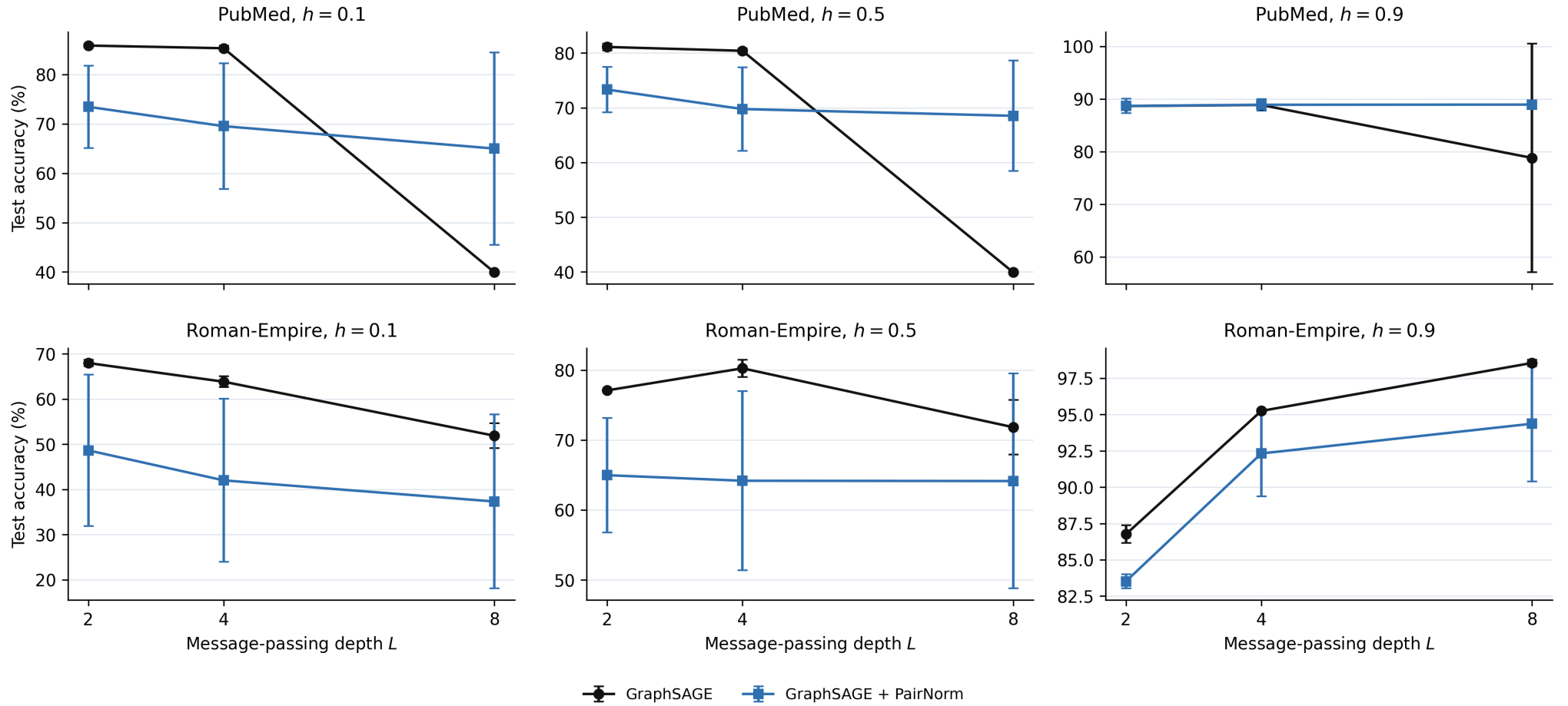


Figure 10: Controlled homophily overview for PubMed-derived and Roman-Empire-derived graphs. Test accuracy is evaluated at validation-selected checkpoints.

Table 6: Controlled-homophily endpoint results for the $L = 8$ low-label setting.

Dataset	h	GS test accuracy	PN test accuracy	PN effect (pp)	GS NDE	PN NDE	GS effective rank	PN effective rank
PubMed	0.1	39.94%	83.49%	+43.55	5.20×10^{-26}	2.652	1.000	4.688
PubMed	0.5	39.94%	78.09%	+38.16	2.79×10^{-20}	1.404	1.000	4.814
PubMed	0.9	78.80%	88.56%	+9.76	0.134	0.231	2.534	3.476
Roman-Empire	0.1	51.89%	55.56%	+3.67	1.211	1.685	11.278	16.968
Roman-Empire	0.5	71.82%	78.68%	+6.85	0.284	0.726	11.579	20.027
Roman-Empire	0.9	98.55%	98.13%	-0.42	0.082	0.187	9.969	17.178

Note. h = target raw edge homophily; GS = GraphSAGE; PN = GraphSAGE+PairNorm; pp = percentage points. PN effect is PN test accuracy minus GS test accuracy. Accuracy is measured at the best-validation checkpoint; normalized Dirichlet energy (NDE) and effective rank are measured at epoch 2400.

The strongest current contrast is controlled PubMed at low homophily, where baseline GraphSAGE collapses both geometrically and predictively while PairNorm preserves representation diversity and strongly improves accuracy. Roman-Empire is less consistent and includes a high-homophily condition where PairNorm changes the diagnostics without improving accuracy.

6.3 Temporal Dynamics

Endpoint values alone can hide whether a model collapses persistently or recovers later during training. Figure 11 therefore tracks NDE and effective-rank ratio across epochs in contrasting controlled conditions.

Temporal Representation Dynamics

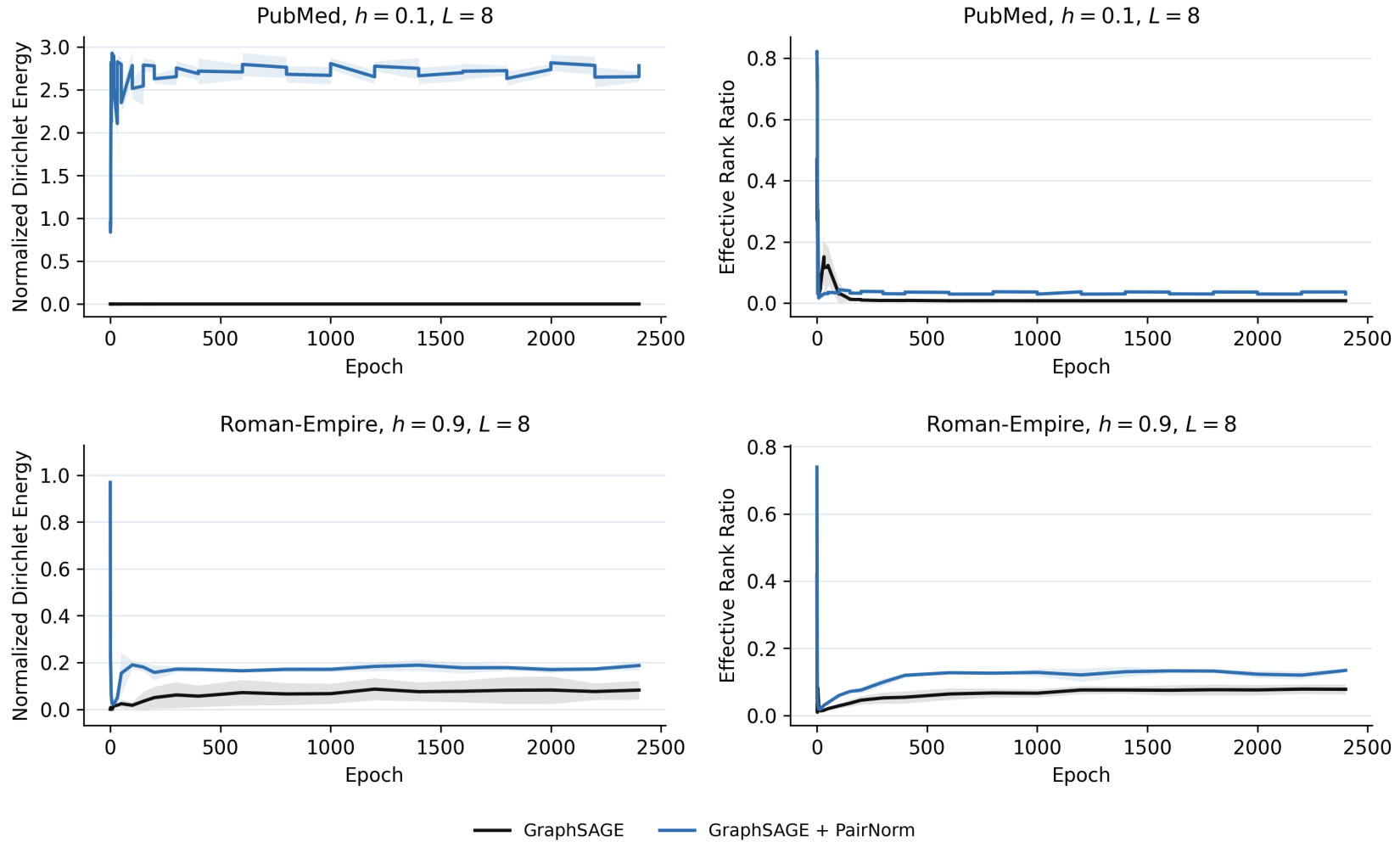


Figure 11: Temporal representation dynamics for contrasting controlled cases. The panels show normalized Dirichlet energy and effective-rank ratio over training epochs, allowing persistent collapse to be distinguished from recovery.

7 Controlled cSBM Experiments

A fully synthetic contextual stochastic block model provides an additional controlled environment [De18]. The thesis-specific parameterization uses 2,000 nodes, two balanced classes, 64 features, target average degree 20, target homophily $\{0.1, 0.5, 0.9\}$, and feature signal $\{0.25, 0.50\}$. Features are isotropic Gaussian noise plus a class-dependent shift along one random unit direction. Five coupled graph/training seeds are used.

Table 7: Controlled cSBM experimental design.

Factor	Setting
Graph family	Contextual stochastic block model
Nodes	2,000
Node features	64
Classes	2 balanced classes
Target homophily	0.1, 0.5, 0.9
Target average degree	20
Feature signal	0.25, 0.50
Feature construction	Isotropic Gaussian noise + class-dependent shift along one random unit direction
Class-mean separation	Equal to feature-signal parameter
Models	GraphSAGE; GraphSAGE + PairNorm
Depths	2, 7, 8
Hidden dimension	128
Train / validation / test	5% / 10% / 85%
Training nodes	100 total (50 per class)
Validation nodes	200 total (100 per class)
Test nodes	1,700 total (850 per class)
Available mask splits	10 stratified splits per graph
Evaluated split	Split 0
Replications	5 coupled graph/training seeds
Seeds	1, 2, 3, 4, 5
Learning rate	0.01
Training epochs	200
Dropout	0.5
Weight decay	0.0005
Total runs	180

7.1 Results

The $L = 8$ cSBM comparison checks whether the strongest controlled PairNorm effects can be reproduced in a fully synthetic setting and how they depend on the feature signal.

Controlled cSBM Performance at $L = 8$

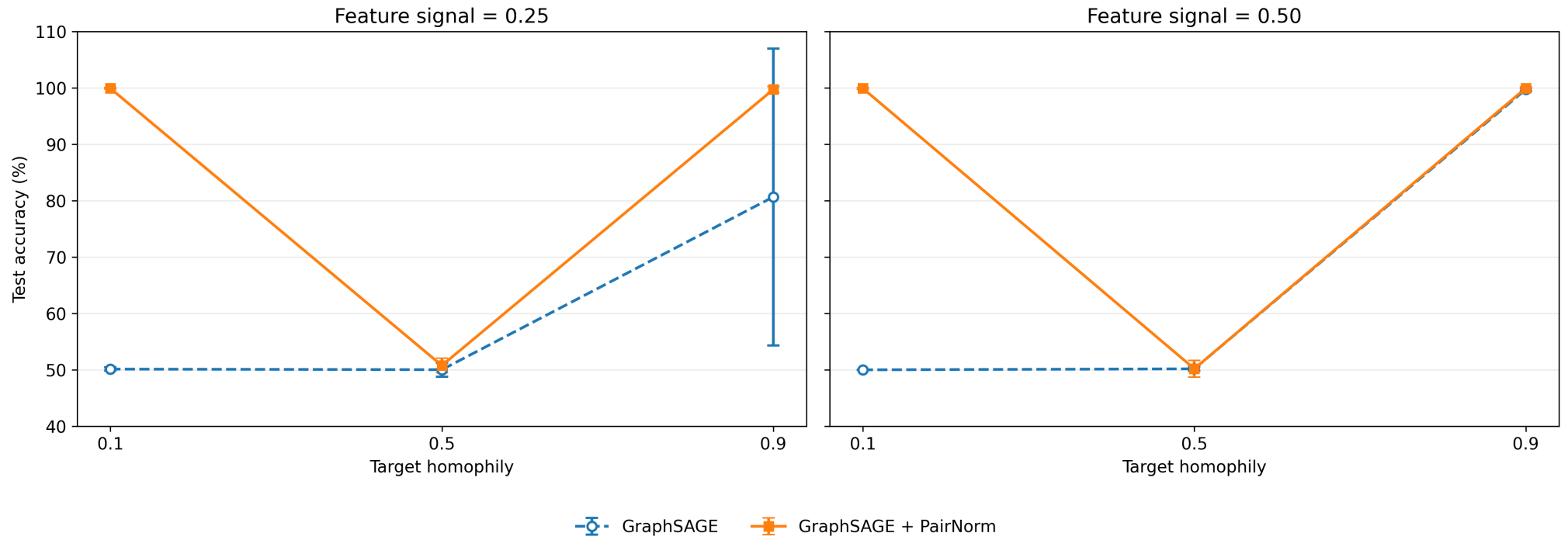


Figure 12: Test accuracy of eight-layer GraphSAGE with and without PairNorm on controlled cSBM graphs across target homophily levels and two feature-signal strengths. Points show means across five coupled graph/training seeds; error bars indicate standard deviations.

The synthetic results partially reproduce strong PairNorm gains, but the effect depends strongly on the graph/feature condition. This supports a conditional interpretation rather than a claim that PairNorm universally fixes deep-message-passing degradation.

8 Learning-Rate Sensitivity

Optimization sensitivity is tested explicitly on controlled $h = 0.1$ PubMed and Roman-Empire graphs with GraphSAGE and GraphSAGE+PairNorm at $L = 8$, hidden dimension 128, and learning rates $\{0.001, 0.003, 0.01, 0.03\}$. Each condition contains five graph seeds and five initialization seeds. Initializations are first averaged within graph seed, followed by summarization across the five graph-level means.

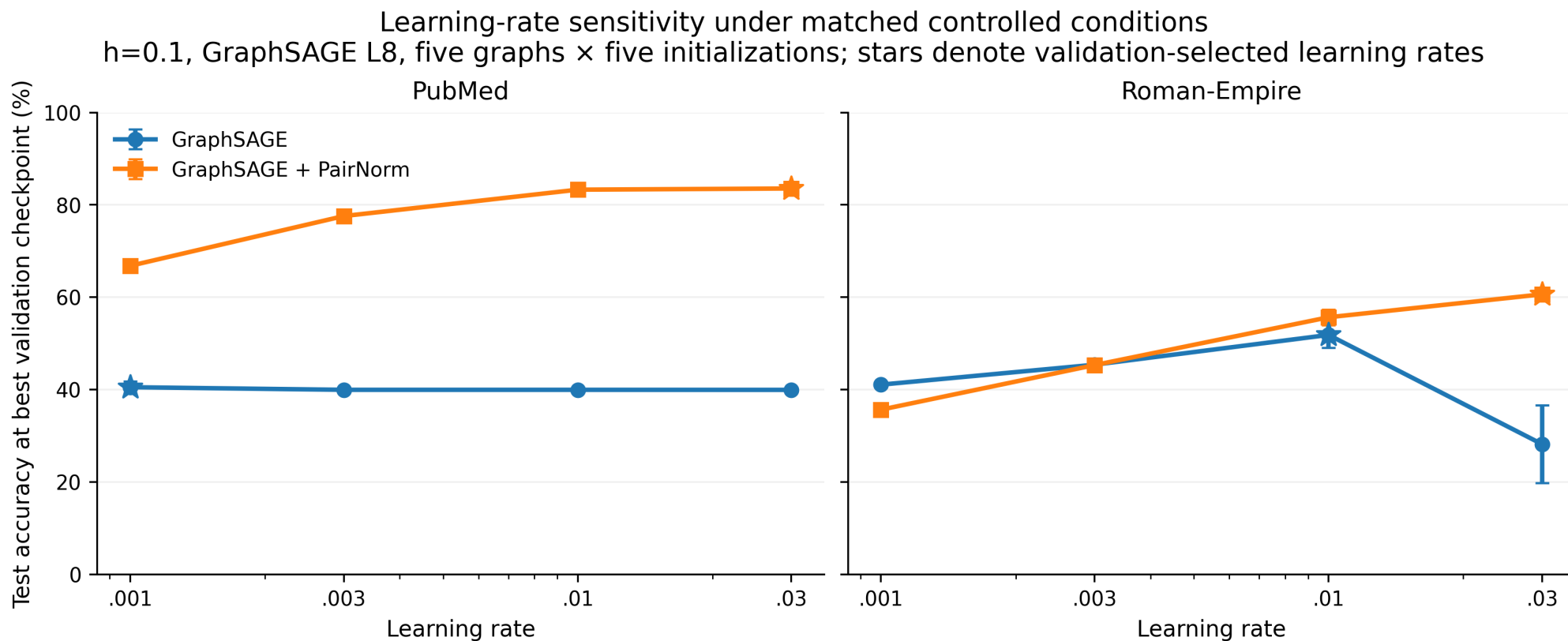


Figure 13: Learning-rate sensitivity of eight-layer GraphSAGE with and without PairNorm on controlled $h = 0.1$ PubMed and Roman-Empire graphs. Error bars summarize graph-level means; stars indicate learning rates selected exclusively by mean validation accuracy.

The current interpretation is intentionally restricted to the tested grid: PubMed GraphSAGE remains near collapse throughout that range, whereas Roman-Empire is substantially more learning-rate sensitive. This separates a robust collapse case from a case where optimization is an important competing explanation.

9 PairNorm Component Ablation

The component ablation separates four variants: baseline, center only, scale only, and full PairNorm. Center-only and scale-only are thesis-specific diagnostic variants. The controlled conditions are PubMed $h = 0.1$ at learning rate 0.01, Roman-Empire $h = 0.9$ at learning rate 0.01, and cSBM $h = 0.1$, feature signal 0.50 at learning rate 0.03. Each condition uses five graph seeds and five initialization seeds.

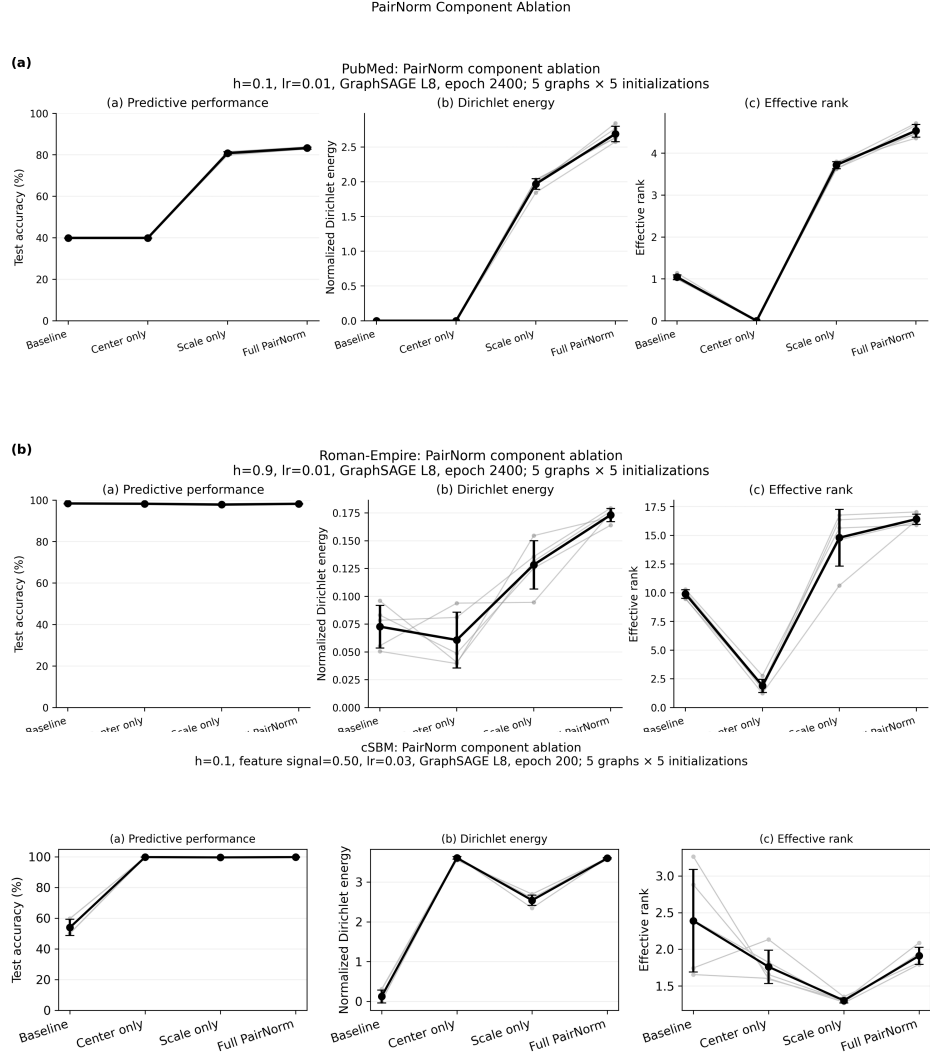


Figure 14: PairNorm component ablations for controlled PubMed, Roman-Empire, and cSBM conditions. Results are first averaged across five model initializations within each graph seed; black markers and error bars show the mean and standard deviation across five graph seeds, while gray lines connect the four variants within each graph seed.

Table 8: PairNorm component-ablation results.

Condition	Learning rate	Variant	Test accuracy	NDE	Effective rank
PubMed (h = 0.1)	0.01	Baseline	39.94% $\pm$ 0.00	5.34e-16 $\pm$ 2.61e-15	1.043 $\pm$ 0.145
PubMed (h = 0.1)	0.01	Center only	39.94% $\pm$ 0.00	6.17e-31 $\pm$ 2.83e-30	0.00e+00 $\pm$ 0.00e+00
PubMed (h = 0.1)	0.01	Scale only	80.82% $\pm$ 1.00	1.967 $\pm$ 0.121	3.719 $\pm$ 0.214
PubMed (h = 0.1)	0.01	Full PairNorm	83.28% $\pm$ 0.52	2.689 $\pm$ 0.149	4.539 $\pm$ 0.211
Roman-Empire (h = 0.9)	0.01	Baseline	98.38% $\pm$ 0.29	0.073 $\pm$ 0.028	9.895 $\pm$ 0.763
Roman-Empire (h = 0.9)	0.01	Center only	98.23% $\pm$ 0.37	0.061 $\pm$ 0.043	1.894 $\pm$ 0.927
Roman-Empire (h = 0.9)	0.01	Scale only	97.86% $\pm$ 0.35	0.128 $\pm$ 0.030	14.796 $\pm$ 3.171
Roman-Empire (h = 0.9)	0.01	Full PairNorm	98.21% $\pm$ 0.22	0.173 $\pm$ 0.017	16.413 $\pm$ 1.117
cSBM (h = 0.1, FS = 0.50)	0.03	Baseline	54.00% $\pm$ 13.73	0.120 $\pm$ 0.415	2.389 $\pm$ 2.111
cSBM (h = 0.1, FS = 0.50)	0.03	Center only	99.87% $\pm$ 0.20	3.610 $\pm$ 0.049	1.760 $\pm$ 0.351
cSBM (h = 0.1, FS = 0.50)	0.03	Scale only	99.68% $\pm$ 0.42	2.542 $\pm$ 0.203	1.301 $\pm$ 0.061
cSBM (h = 0.1, FS = 0.50)	0.03	Full PairNorm	99.87% $\pm$ 0.24	3.600 $\pm$ 0.038	1.910 $\pm$ 0.308

Note. h = target homophily; FS = feature signal; NDE = normalized Dirichlet energy. Values are mean $\pm$ SD across the 25 individual runs. The table summarizes the same run sets as Figure 14, whereas Figure 14 first averages initializations within graph seed.

The component behavior is domain dependent. In PubMed, scaling is the dominant successful component and centering alone does not recover performance. In the cSBM condition, centering alone is sufficient for near-perfect accuracy, while scaling alone also performs strongly. Roman-Empire again shows that substantial geometric changes need not translate into a useful accuracy improvement.

10 Oversmoothing Synthesis

This section is intended to answer RQ2 directly. The current evidence supports the following thesis-level synthesis:

1. Depth degradation is heterogeneous across datasets and controlled graph conditions.
2. PairNorm can reduce or increase depth degradation depending on the condition.
3. Persistent representation collapse can coincide with severe predictive failure, most clearly in controlled PubMed.
4. Higher NDE or effective rank does not guarantee higher predictive accuracy.
5. Homophily modulates the phenomenon but does not determine it by itself.
6. Some apparent depth failures are strongly optimization sensitive, while others persist across the tested learning-rate range.
7. The PairNorm component responsible for a successful intervention is domain dependent.

11 Oversquashing

This section introduces the theoretical motivation and experimental logic for the oversquashing analysis. It provides the basis for the empirical studies in Sections 12–14 and the synthesis in Section 15. Across the three empirical oversquashing studies, the analysis contains 1,860 training runs: 1,200 controlled bottleneck runs, 240 direct-sensitivity runs, and 420 real-world rewiring runs including the final budget-robustness extension.

Oversquashing concerns the difficulty of preserving and transmitting long-range information when many signals must pass through restricted graph structure and finite-dimensional representations. Prior work connects this problem to graph bottlenecks, curvature, source-target distance, width, depth, and topology [AY21; Di23; To22]. In this thesis, oversquashing is therefore not diagnosed from accuracy degradation alone. The empirical logic combines controlled structural variation, direct source-target sensitivity, and real-world structural interventions.

The schematic below is conceptual only. The crossed-ring construction is thesis-specific: it adds opposite-node shortcuts and therefore changes both connectivity and source-target distance. It must not be interpreted as isolating bottleneck width at fixed distance.

Graph Structure and Long-Range Information Flow

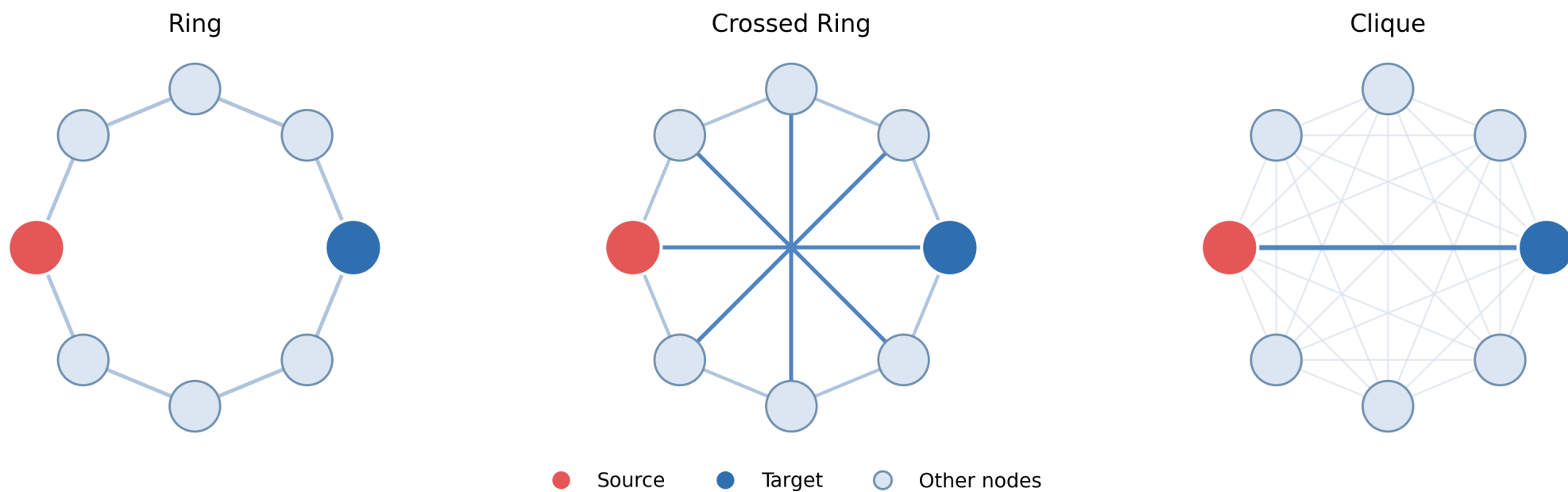


Figure 15: Schematic graph structures illustrating how connectivity can alter long-range information pathways between the same source and target nodes. The ring provides sparse multi-hop routes, the thesis-specific crossed-ring construction adds opposite-node shortcuts, and the clique provides dense connectivity. This is a conceptual figure, not an empirical result.

12 Controlled Long-Range Bottleneck Benchmark

The controlled benchmark varies source-target distance $d \in \{2, 4, 6, 8, 10\}$ and bottleneck width $B \in \{1, 2, 4, 8\}$. The number of GraphSAGE message-passing layers equals the source-target distance. Four variants are compared: GraphSAGE (GS), GraphSAGE+PairNorm (PN), rewired GraphSAGE (RW), and PairNorm+rewiring (PN+RW). Each structural condition uses five graph seeds and three initialization seeds, yielding 1,200 runs in total.

The strongest causal structural comparison is width at fixed distance and fixed depth. For $d \geq 4$, baseline GraphSAGE accuracy increases monotonically with bottleneck width. Distance itself is not isolated causally because distance and depth are coupled in this benchmark. Rewiring is a strong positive structural intervention, but it reduces the source-target distance to one and therefore does not isolate bottleneck width at fixed distance.

Controlled long-range bottleneck benchmark

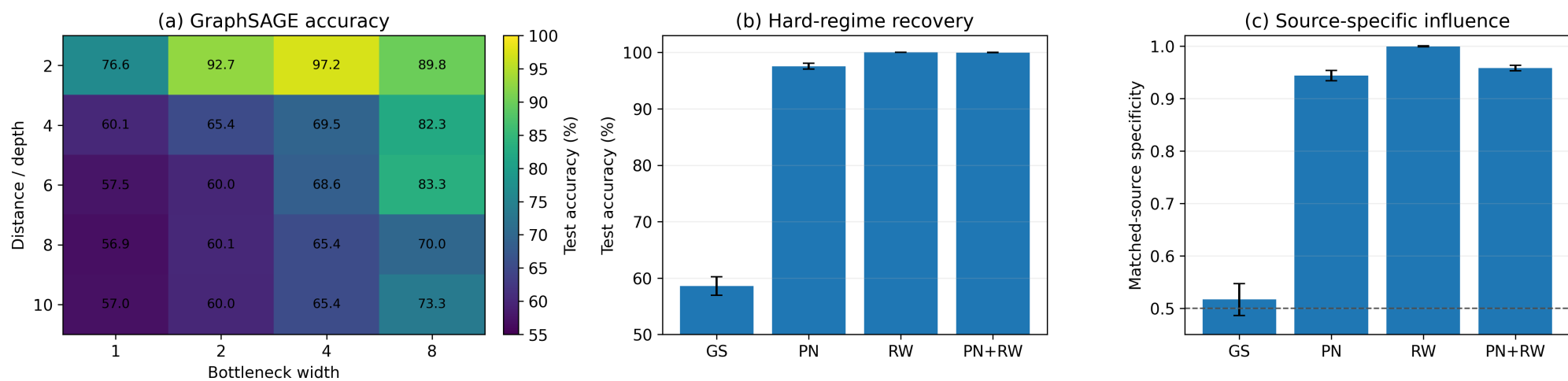


Figure 16: Controlled synthetic bottleneck benchmark. Panel (a) reports mean baseline GraphSAGE test accuracy over 15 runs per distance-width cell. Panels (b) and (c) summarize the difficult long-range/narrow-bottleneck regime ($d \geq 6$, $B \leq 2$). Specificity is matched-source influence divided by matched plus distractor influence; values near 0.5 indicate that the target is no more sensitive to the matched source than to the distractor.

In the hard regime, baseline GraphSAGE reaches only $58.59 \pm 1.64\%$ accuracy and matched-source specificity 0.5166 ± 0.0304 . PairNorm raises these values to $97.56 \pm 0.53\%$ and 0.9438 ± 0.0099 , while rewiring reaches $100.00 \pm 0.00\%$ and 0.9994 ± 0.0009 . The combined PN+RW variant reaches $99.99 \pm 0.03\%$ and 0.9579 ± 0.0053 . PairNorm therefore recovers the controlled task almost as strongly as the structural intervention, so the failure should not be described as exclusively structural oversquashing.

Table 9: Hard-regime summary ($d \geq 6$, $B \leq 2$). Values are graph-level mean $\pm$ SD after averaging initialization seeds within each graph seed.

Variant	Test accuracy (%)	Matched-source specificity
GraphSAGE	58.59 ± 1.64	0.5166 ± 0.0304
GraphSAGE + PairNorm	97.56 ± 0.53	0.9438 ± 0.0099
Rewired GraphSAGE	100.00 ± 0.00	0.9994 ± 0.0009
PairNorm + rewiring	99.99 ± 0.03	0.9579 ± 0.0053

13 Direct Source-Target Sensitivity

This section evaluates four representative structural cells: $(2, 1)$, $(2, 8)$, $(8, 1)$, and $(8, 8)$, where each pair denotes (d, B) . The same four model variants are evaluated with five graph seeds and three initialization seeds, giving 240 runs. Sensitivity is measured after loading the best-validation checkpoint. The primary direct diagnostics are (i) a finite-difference source-bit intervention and (ii) a signal-coordinate Jacobian measuring the derivative of the target logits with respect to the actual source-signal feature. A full-source Frobenius Jacobian is retained as a supplementary diagnostic.

Direct source-target sensitivity diagnostics

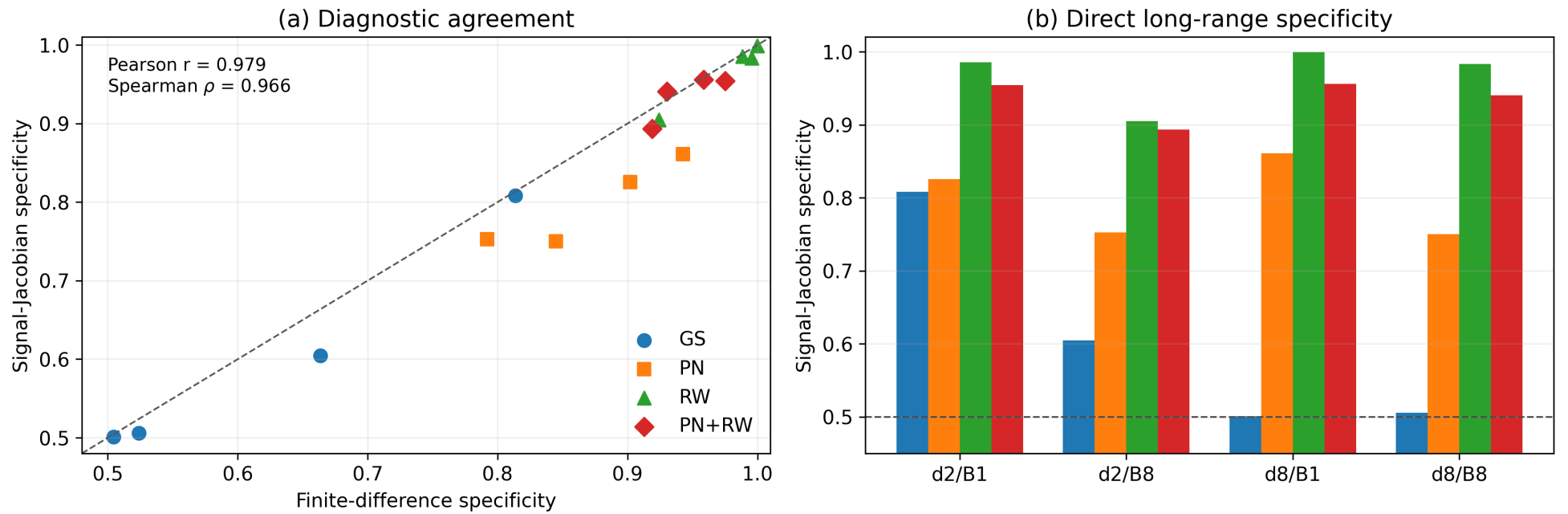


Figure 17: Direct source-target sensitivity. Panel (a) compares finite-difference and signal-coordinate Jacobian specificity across the 16 model/structure condition means. Panel (b) shows signal-Jacobian specificity in the four selected structural cells. The two direct diagnostics agree strongly overall (Pearson $r = 0.9786$, Spearman $\rho = 0.9658$).

The finite-difference and signal-Jacobian diagnostics agree strongly, which supports the finite-difference measure as a direct long-range diagnostic. At the same time, sensitivity is not a universal proxy for predictive accuracy. For baseline GraphSAGE at $d = 8$, widening the bottleneck from $B = 1$ to $B = 8$ increases accuracy by 16.01 percentage points, while specificity changes only slightly. For PairNorm, the same width change increases accuracy from 97.09% to 99.49% even though finite-difference and signal-Jacobian specificity both decrease. The full Frobenius specificity remains near 0.5 for non-rewired GraphSAGE and PairNorm and is therefore treated as a negative/supplementary diagnostic rather than as the primary measure.

Table 10: Direct-sensitivity condition means. FD = finite-difference specificity; Signal-Jac = signal-coordinate Jacobian specificity; Frob-Jac = full-source Frobenius Jacobian specificity. Each row summarizes 15 runs.

d	B	Variant	Accuracy (%)	FD	Signal-Jac	Frob-Jac
2	1	GS	76.40	0.8137	0.8081	0.5000
2	1	PN	96.18	0.9019	0.8257	0.4983
2	1	RW	100.00	0.9883	0.9855	0.9860
2	1	PN+RW	100.00	0.9750	0.9543	0.9658
2	8	GS	89.88	0.6635	0.6048	0.5000
2	8	PN	99.94	0.7920	0.7527	0.5002
2	8	RW	100.00	0.9240	0.9049	0.9035
2	8	PN+RW	100.00	0.9190	0.8935	0.8914
8	1	GS	56.67	0.5045	0.5013	0.5000
8	1	PN	97.09	0.9425	0.8610	0.5095
8	1	RW	100.00	0.9998	0.9992	0.9994
8	1	PN+RW	99.98	0.9586	0.9561	0.9578
8	8	GS	72.68	0.5239	0.5057	0.5000
8	8	PN	99.49	0.8447	0.7501	0.5003
8	8	RW	100.00	0.9956	0.9831	0.9870
8	8	PN+RW	100.00	0.9303	0.9406	0.9330

14 Real-World Structural Intervention and Budget Robustness

The real-world transfer experiment uses the natural PubMed and Roman-Empire topologies together with low-label splits. A Fiedler/Cheeger sweep identifies a pre-defined bottleneck cut on each natural graph. Targeted rewiring adds edges across this cut. The matched random control uses exactly the same selected endpoints and the same number of added edges, but pairs the endpoints randomly. Consequently, targeted and random interventions have the same added degree for every node while differing in how strongly they bypass the pre-defined cut. Construction does not use labels or features; labels are used only afterward to record homophily.

The original 1% edge-budget experiment was extended to 0.5% and 2% budgets to test whether the result depended on a single intervention strength. The resulting 420 training runs comprise two datasets, GraphSAGE and GraphSAGE+PairNorm, five paired replicate seeds, three initialization seeds, and the original/random/targeted conditions required by the three-budget analysis. The five replicate seeds correspond to split/intervention pairings on the same natural base topology; they are not five independent natural graph draws. Initialization seeds are averaged first, leaving five paired replicate-level observations for the primary contrasts.

Real-world rewiring: accuracy across intervention budgets

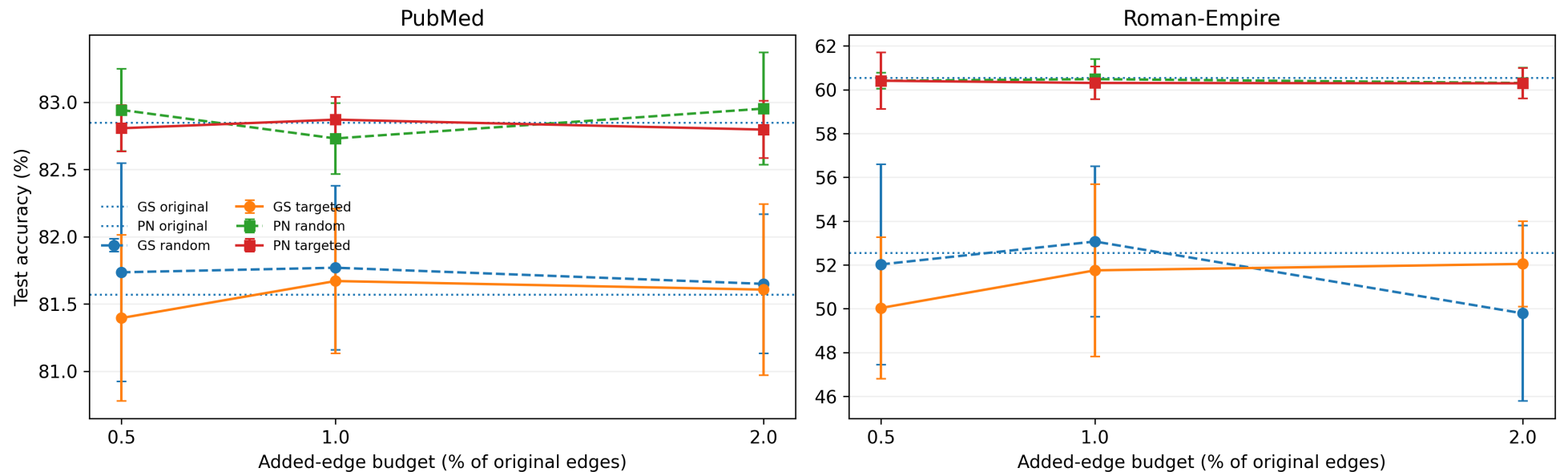


Figure 18: Real-world test accuracy across 0.5%, 1%, and 2% added-edge budgets. Error bars show SD across five replicate-level means after averaging the three initialization seeds. Dotted horizontal lines mark the corresponding original-topology baselines.

Targeted versus degree-matched random rewiring

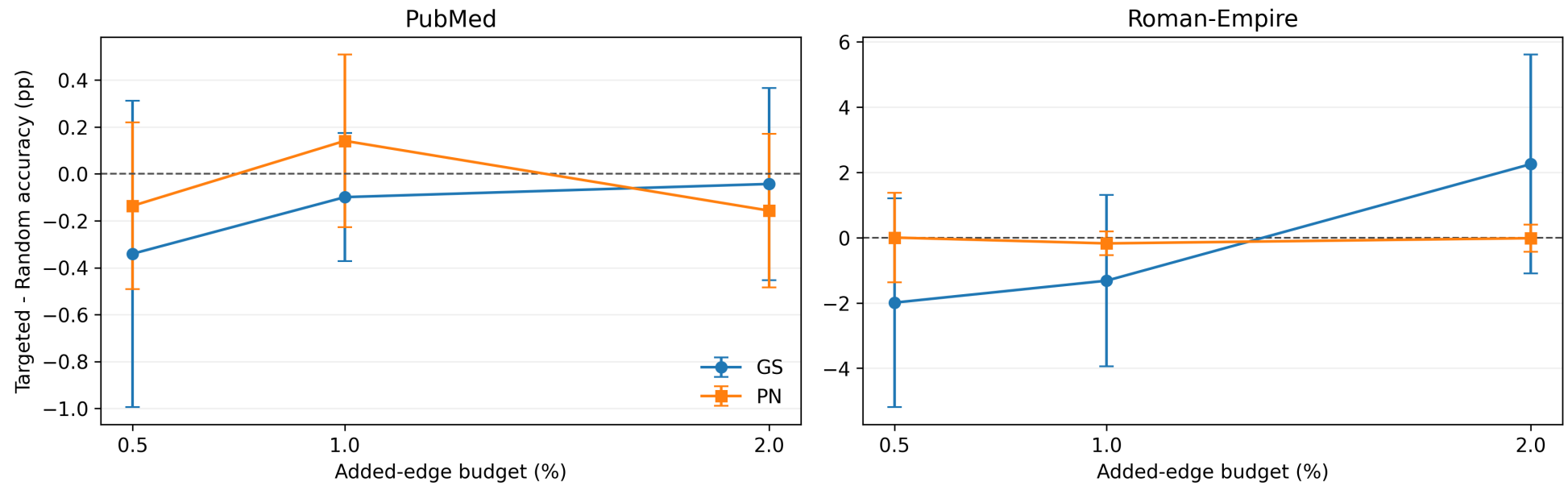


Figure 19: Primary real-world structural control: targeted minus degree-matched random test accuracy across edge budgets. Error bars are 95% confidence intervals over the five paired replicate-level differences. All shown intervals include zero.

Strength of the pre-identified bottleneck-cut intervention

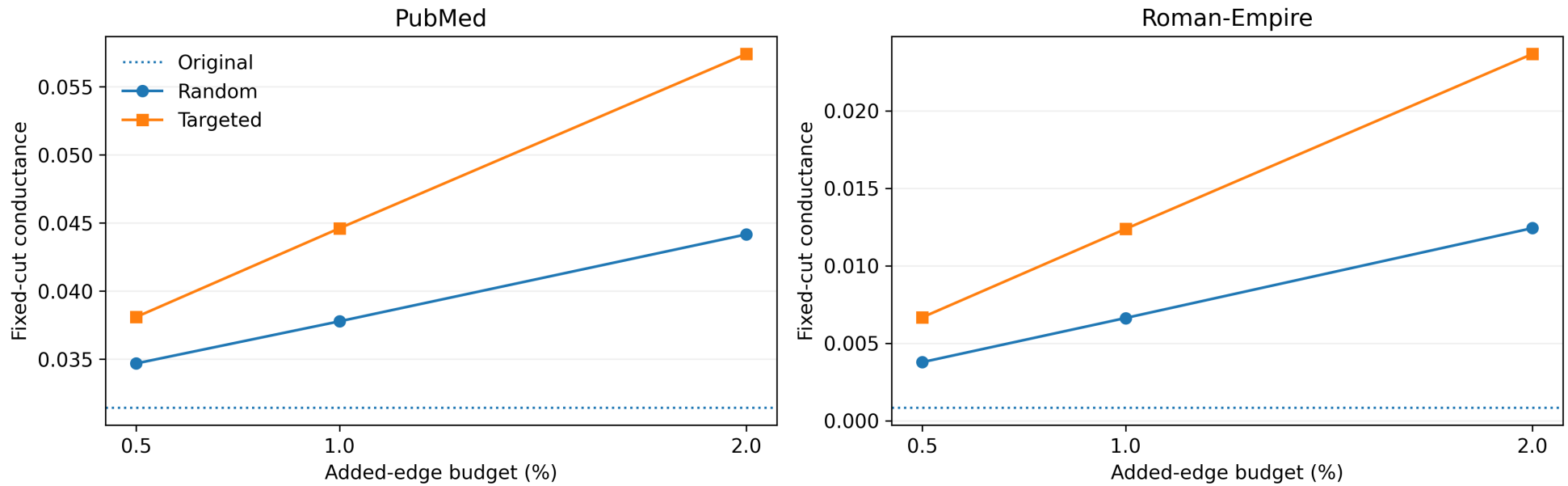


Figure 20: Structural intervention strength across budgets. Targeted rewiring increases the conductance of the pre-identified cut more strongly than the degree-matched random control at every tested budget, while Figure 19 shows no corresponding consistent predictive advantage.

The structural manipulation is therefore successful by construction and by the measured fixed-cut conductance. For PubMed, targeted conductance rises from 0.03810 at 0.5% to 0.05741 at 2%; for Roman-Empire it rises from 0.00666 to 0.02367. However, the predictive response does not track this monotonic structural improvement. The targeted-minus-random effect changes in magnitude and, for Roman-Empire GraphSAGE, even changes sign across budgets. PairNorm shows no consistent additional synergy with targeted rewiring.

Table 11: Primary paired contrast, targeted minus degree-matched random accuracy. Effects are in percentage points. Confidence intervals and exact sign-flip tests use $n = 5$ paired replicate-level differences.

Dataset	Model	Budget	Mean (pp)	95% CI	+/- replicates	Exact p
PubMed	GS	0.5%	-0.3405	[-0.9935, 0.3125]	2 / 3	0.2500
PubMed	GS	1.0%	-0.0991	[-0.3717, 0.1735]	1 / 4	0.3750
PubMed	GS	2.0%	-0.0430	[-0.4526, 0.3666]	3 / 2	0.8750
PubMed	PN	0.5%	-0.1360	[-0.4913, 0.2192]	2 / 3	0.3750
PubMed	PN	1.0%	0.1408	[-0.2270, 0.5086]	3 / 2	0.3750
PubMed	PN	2.0%	-0.1567	[-0.4847, 0.1712]	1 / 4	0.3125
Roman-Empire	GS	0.5%	-1.9900	[-5.1867, 1.2067]	1 / 4	0.1875
Roman-Empire	GS	1.0%	-1.3169	[-3.9400, 1.3062]	1 / 4	0.2500
Roman-Empire	GS	2.0%	2.2575	[-1.0980, 5.6131]	3 / 2	0.2500
Roman-Empire	PN	0.5%	0.0042	[-1.3741, 1.3824]	3 / 2	1.0000
Roman-Empire	PN	1.0%	-0.1748	[-0.5366, 0.1871]	2 / 3	0.2500
Roman-Empire	PN	2.0%	-0.0149	[-0.4358, 0.4060]	3 / 2	0.9375

These data do not justify a categorical null claim. The appropriate conclusion is that there is no consistent evidence, under the tested 0.5–2% intervention budgets, that stronger targeted bypass of the pre-identified bottleneck cut improves node-classification accuracy relative to the original topology or to the degree-matched random control. PubMed homophily decreases slightly as the rewiring budget grows, while Roman-Empire homophily changes only minimally; this accompanying change is recorded as a caveat rather than treated as the causal explanation.

15 Oversquashing Synthesis

This section is intended to answer RQ3 directly. The current evidence supports the following thesis-level synthesis:

1. In the controlled synthetic task, long-range degradation is strongly bottleneck-sensitive: at fixed distance and depth, wider bottlenecks improve baseline GraphSAGE accuracy, especially for $d \geq 4$.
2. In difficult long-range/narrow-bottleneck regimes, baseline matched-source specificity approaches 0.5, while both PairNorm and structural rewiring can restore predictive performance and source-specific influence.
3. Finite-difference and signal-coordinate Jacobian diagnostics agree strongly, but source-target sensitivity is not a universal proxy for predictive accuracy; the full Frobenius Jacobian is particularly weak for non-rewired GS/PN conditions.
4. On natural PubMed and Roman-Empire graphs, targeted rewiring causally strengthens the intended pre-defined bottleneck cut more than a degree-matched random control across three intervention budgets, yet the stronger structural change does not consistently improve node-classification accuracy.
5. Oversquashing is therefore a useful conditional mechanism for deep-GNN failure in tasks that genuinely require distant information transfer, but improving a bottleneck proxy alone is not sufficient to guarantee better downstream performance. PairNorm’s benefits also cannot be reduced to the same mechanism as explicit structural bottleneck relief.

16 Discussion

This chapter is intended to discuss the overall findings of the thesis in relation to the research questions and the relevant literature.

17 Conclusion

This chapter is intended to summarize the main findings and answer the research questions.

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